The first-moment Berndt integral of order three
MathDB 360892

Research note

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Task 1. Prove the exact integral evaluation

Write $G = \Gamma(1/4)$. We prove Conjecture 4.3 of Rui–Xu–Zhao [1, 2], with precisely its stated integrand and constants. Theorem 4.1 of those sources treats numerators x^{4m+1} for $m > 0$; the present numerator x is outside that range.

Theorem 1.

$$\boxed{\int_0^\infty \frac{x\,dx}{(\cos x + \cosh x)^3} = -\frac{G^4}{2^7\pi^2} + \frac{G^4}{2^9\pi} + \frac{G^{12}}{2^{13}\pi^7}.}$$

1. A convergent residue formula

Denote the integral by I, and put

$$a = \frac{1+i}{2}, \qquad b = \frac{1-i}{2}, \qquad t_n = \pi(n+\tfrac{1}{2}), \qquad h(t) = \operatorname{sech}^3 t.$$

The integral converges: the integrand is $O(x)$ at zero and $O(xe^{-3x})$ at infinity. Use

$$F(z) = \frac{z}{(\cos z + \cosh z)^3}, \qquad \cos z + \cosh z = 2\cosh(az)\cosh(bz).$$

On the positively oriented triangle with vertices $0, R, iR$, take $R = 2\pi N$, N a positive integer. For the oblique edge $x + y = R$, $\operatorname{Im}(az) = \pi N$ and $\operatorname{Re}(bz) = R/2$. Consequently $|\cosh(az)| \geq 1$ and $|\cosh(bz)| \geq \sinh(R/2)$. Its integral is therefore $O(R^2 e^{-3R/2})$. Also $F(iy) = iF(y)$, so the two axial integrals sum to $2\int_0^R F(x)\,dx$. The first-quadrant poles are exactly $z_n = (1+i)t_n$, of order three. Thus

$$I = \pi i \sum_{n\geq 0} \operatorname{Res}_{z=z_n} F(z). \tag{1}$$

To evaluate a residue, write $z = z_n + v$. Then

$$F(z_n+v) = \frac{i(-1)^n}{8a^3v^3}(z_n+v)h(t_n+bv)\left(\frac{av}{\sinh(av)}\right)^3, \qquad \left(\frac{av}{\sinh(av)}\right)^3 = 1 - \frac{a^2v^2}{2} + O(v^4).$$

The coefficient of v^{-1}, multiplied by πi, is

$$\pi i \operatorname{Res}_{z=z_n} F(z) = \frac{\pi(-1)^n}{8}\{t_n h''(t_n) + 2h'(t_n) + t_n h(t_n)\}.$$

These terms decay exponentially, which also justifies the limiting sum in (1). If $s(t) = \operatorname{sech} t$, then $s'' = s - 2s^3$. Replacing h by $(s - s'')/2$ gives the particularly useful formula

$$I = \frac{\pi}{16}\sum_{n\geq 0}(-1)^n\{t_n(s - s^{(4)})(t_n) + 2(s' - s^{(3)})(t_n)\}. \tag{2}$$

2. Interchange the divisor variables

For nonnegative integers P, Q and $|q| < 1$, define

$$\Phi_{P,Q}(q) = \sum_{k,v\ge 1} k^P v^Q q^{kv}, \qquad q_* = e^{-\pi/2}, \qquad T_{P,Q} = \frac{\Phi_{P,Q}(iq_*) - \Phi_{P,Q}(-iq_*)}{2i}.$$

Let $\chi_4(j)$ be zero for even j and $(-1)^{(j-1)/2}$ for odd j. Since $\chi_4(kv) = \chi_4(k)\chi_4(v)$, we have

$$T_{P,Q} = \sum_{k,v\ge 1} \chi_4(k)\chi_4(v) k^P v^Q q_*^{kv}.$$

The absolutely convergent expansion $s(t) = 2\sum_{v\ge 1} \chi_4(v) e^{-vt}$ now yields, for every nonnegative derivative order d,

$$\sum_{n\ge 0} (-1)^n t_n^P s^{(d)}(t_n) = 2^{1-P}\pi^P (-1)^d T_{P,d}. \tag{3}$$

All rearrangements and differentiations here are justified by absolute convergence; for example $e^{-\pi kv/2}$ dominates every fixed polynomial weight. Applying (3) to (2),

$$I = \frac{\pi^2}{16}(T_{1,0} - T_{1,4}) - \frac{\pi}{4}(T_{0,1} - T_{0,3}). \tag{4}$$

Use the normalization $q = e^{2\pi i\tau}$ and $D = q\,d/dq$. For $r \ge 1$ put $L_r(q) = \sum_{k,v\ge 1} v^r q^{kv} = \sum_{n\ge 1} \sigma_r(n) q^n$. The standard Eisenstein series satisfy

$$E_2 = 1 - 24L_1, \qquad E_4 = 1 + 240L_3, \qquad E_6 = 1 - 504L_5.$$

Interchanging k and v, and differentiating absolutely convergent series, proves

$$\Phi_{1,0} = \Phi_{0,1} = \frac{1 - E_2}{24}, \qquad \Phi_{0,3} = \frac{E_4 - 1}{240}, \tag{5}$$

$$\Phi_{1,4} = DL_3 = \frac{E_2E_4 - E_6}{720}. \tag{6}$$

The last equality is Ramanujan's differential identity $DE_4 = (E_2E_4 - E_6)/3$ [3, Section 5.1]. In particular, no series with a negative power of its divisor variable has to be evaluated.

3. The required modular values

Put

$$u = \frac{G^4}{4\pi^3}, \qquad \tau_+ = \frac{1+i}{4}, \qquad \tau_- = \frac{-1+i}{4}.$$

We will establish

$$E_2(\tau_+) = \frac{12}{\pi} - 3iu, \qquad E_4(\tau_+) = -33u^2, \qquad E_6(\tau_+) = 189iu^3. \tag{7}$$

Here are the details, including the weight-two correction.

Use the standard theta constants

$$\theta_2(\tau) = \sum_{n\in\mathbb{Z}} e^{\pi i\tau(n+1/2)^2}, \quad \theta_3(\tau) = \sum_{n\in\mathbb{Z}} e^{\pi i\tau n^2}, \quad \theta_4(\tau) = \sum_{n\in\mathbb{Z}} (-1)^n e^{\pi i\tau n^2}.$$

The classical values are $E_2(i) = 3/\pi$ and $E_4(i) = 3u^2/4$ [3, p. 85]. Theta transformation and Jacobi's identity give $\theta_2(i)^4 = \theta_4(i)^4 = \theta_3(i)^4/2$. Together with the standard identities [3, pp. 28–29]

$$E_4 = \tfrac{1}{2}(\theta_2^8 + \theta_3^8 + \theta_4^8),$$
$$E_6 = \tfrac{1}{2}(\theta_2^4 + \theta_3^4)(\theta_3^4 + \theta_4^4)(\theta_4^4 - \theta_2^4),$$

and positivity on the imaginary axis, these imply $\theta_3(i)^4 = u$.

The theta duplication identities [4, Section 20.7] are

$$\theta_2(2\tau)^2 = \frac{\theta_3(\tau)^2 - \theta_4(\tau)^2}{2}, \quad \theta_3(2\tau)^2 = \frac{\theta_3(\tau)^2 + \theta_4(\tau)^2}{2}, \quad \theta_4(2\tau)^4 = \theta_3(\tau)^2\theta_4(\tau)^2.$$

Thus, at $2i$,

$$(\theta_2^4, \theta_3^4, \theta_4^4) = u\left(\frac{3-2\sqrt{2}}{8}, \frac{3+2\sqrt{2}}{8}, \frac{1}{\sqrt{2}}\right), \qquad E_4(2i) = \frac{33u^2}{64}, \quad E_6(2i) = \frac{189u^3}{512}.$$

For E_2, the elementary level-two identity is

$$2E_2(2\tau) - E_2(\tau) = \frac{\theta_3(\tau)^4 + \theta_4(\tau)^4}{2}. \tag{8}$$

To see its normalization directly, the coefficient of q^n on the right is the number $r_4(2n)$ of representations as four squares. Jacobi's formula [3, Proposition 11] gives

$$r_4(2n) = 8 \sum_{d \mid 2n,\, 4\nmid d} d = 24\{\sigma_1(n) - 2\sigma_1(n/2)\},$$

where $\sigma_1(n/2) = 0$ for odd n. These are exactly the coefficients on the left, and both constant terms are one. At $\tau = i$, (8) therefore gives $E_2(2i) = 3/(2\pi) + 3u/8$.

Finally set $z = 2i - 2$, so that $\tau_+ = -1/z$. Integer translation leaves the three Eisenstein series unchanged. Their inversion laws are

$$E_2(-1/z) = z^2E_2(z) + \frac{6z}{\pi i}, \qquad E_4(-1/z) = z^4E_4(z), \qquad E_6(-1/z) = z^6E_6(z).$$

Substitution of the three values at $2i$ proves (7).

4. Finish the evaluation

The numbers iq_* and $-iq_*$ correspond to τ_+ and τ_-. The Fourier coefficients of E_2, E_4, E_6 are real, so their values at τ_- are the complex conjugates of those in (7). Consequently (5)–(6) give

$$T_{1,0} = T_{0,1} = \frac{u}{8}, \qquad T_{0,3} = 0, \qquad T_{1,4} = \frac{\mathrm{Im}\{(12/\pi - 3iu)(-33u^2) - 189iu^3\}}{720} = -\frac{u^3}{8}.$$

Formula (4) becomes

$$I = \frac{\pi^2}{128}(u + u^3) - \frac{\pi u}{32}.$$

Replacing u by $G^4/(4\pi^3)$ gives exactly the three terms in the theorem. This is an identity obtained by convergent analytic operations and classical modular identities; no numerical fitting or algebraic-independence assumption is used. □

Task 2. Evaluate all first moments

Extend the residue and divisor-interchange calculation to $\int_0^\infty x(\cos x + \cosh x)^{-M}\,dx$ for every positive integer M. Determine the exact period support and resolve the all-power first-moment conjecture of Xu–Zhao, arXiv:2401.01385v2, Conjecture 5.8, including the smallest powers.

Task 3. Resolve the auxiliary series questions

Evaluate the half-integer hyperbolic sums proposed after Conjecture 4.3 in the Research Square version. Check their printed numerators and denominator powers against the contour formula, and identify which individual evaluations are necessary for that alternative proof.

Task 4. Vary the modular parameter

For $q_* = e^{-y/2}$ with $y > 0$, retain the four twisted sums as functions of y. Classify imaginary quadratic values for which their Eisenstein evaluations lead to explicit gamma-value formulas, and find contour integrals that realize these other parameters.

Task 5. Compute an exact recurrence in the power

Combine the recurrence for $\operatorname{sech}^M t$ with Ramanujan's differential equations to obtain a recurrence for the first-moment coefficients. Determine whether it avoids reconstructing every individual residue and whether it yields useful sign or growth information.

References

[1] H. Rui, C. Xu and J. Zhao, *Berndt-Type Integrals of Order Three and Series Associated with Jacobi Elliptic Functions*, arXiv:2311.15666v2 (28 November 2023), Conjecture 4.3, p. 17.

[2] H. Rui, C. Xu and J. Zhao, same title, Research Square version 1 (22 February 2024), Conjecture 4.3, printed pp. 18–19.

[3] D. Zagier, *Elliptic Modular Forms and Their Applications*, in *The 1-2-3 of Modular Forms*, Springer (2008), pp. 1–103; author-hosted chapter.

[4] NIST Digital Library of Mathematical Functions, Section 20.7, theta identities.

Verification and status. The accompanying `verify_evaluation.py` checks the exact residue algebra, modular values, and final constants. Independent numerical diagnostics are reported separately. Both complete source versions and dated public-claim checks accompany the note. The authoring agent performed the audit; independent peer review has not occurred. Tasks 2–5 are proposed directions, not additional solutions.